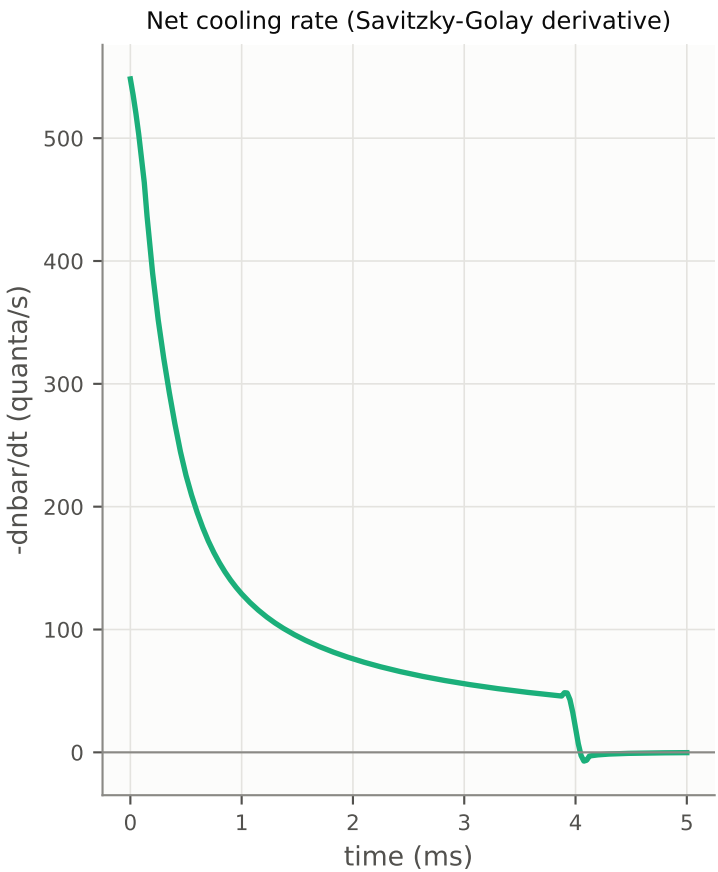
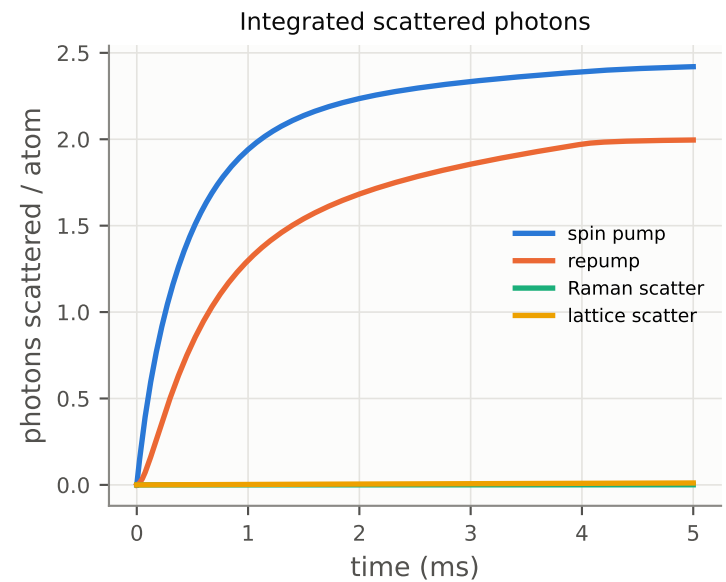
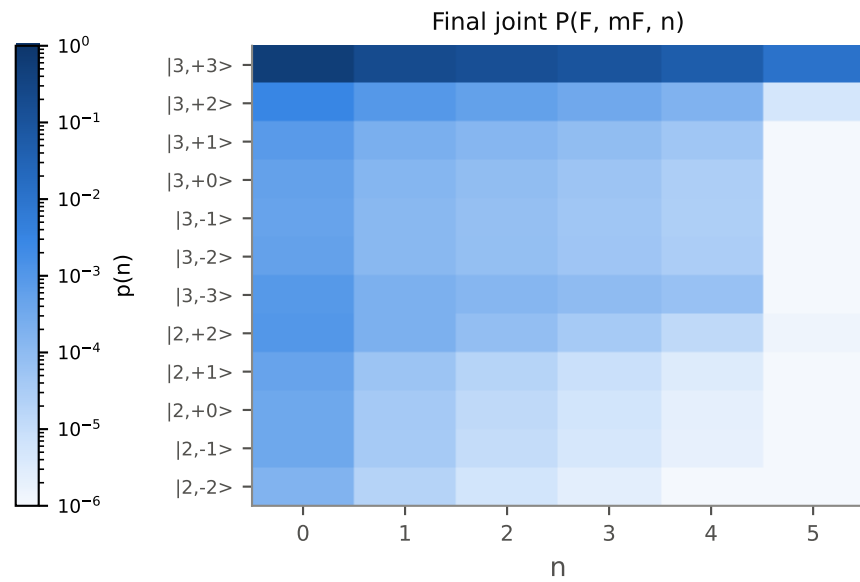
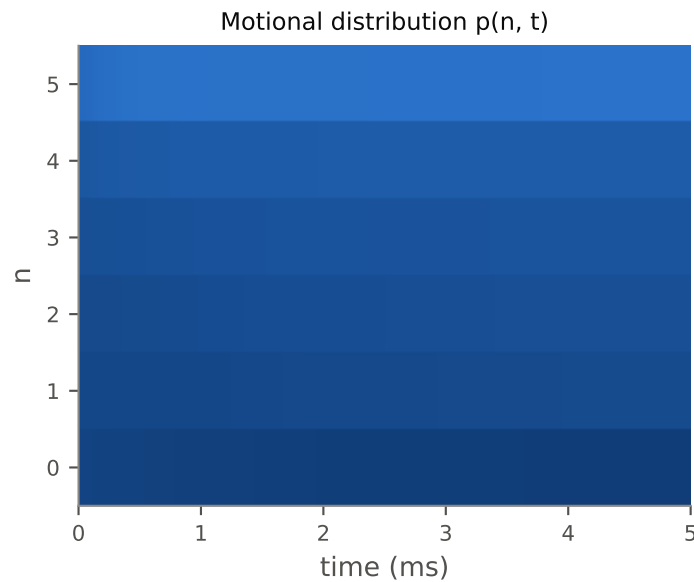
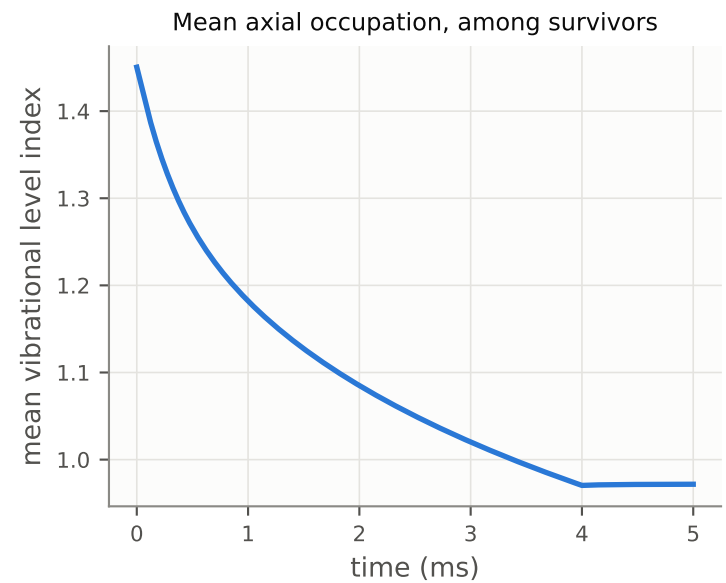
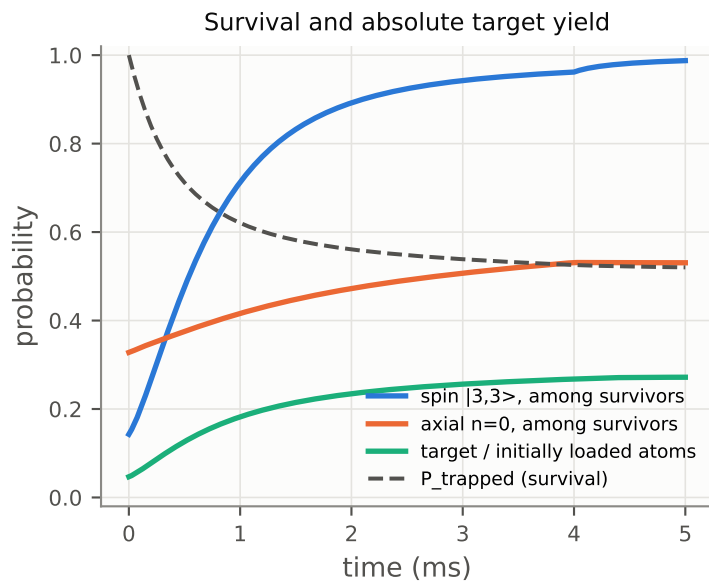
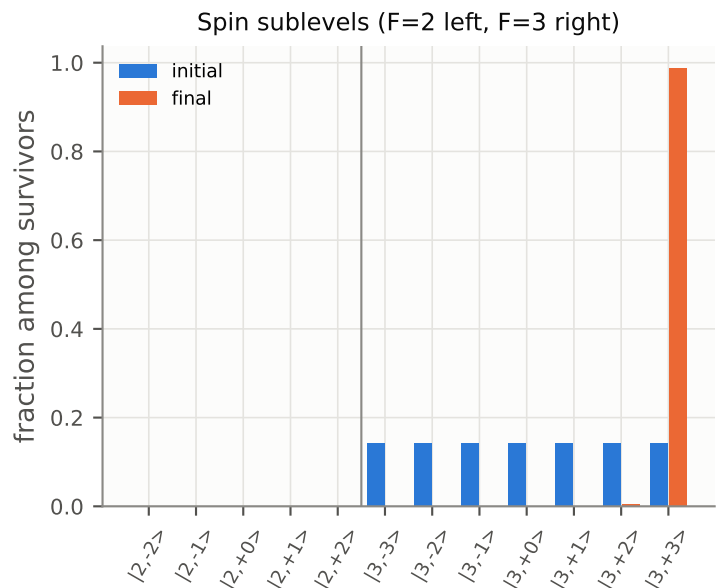
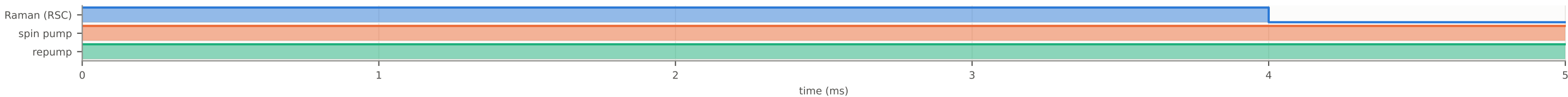


clarified_3d_experiment - clarified 3D experiment

Beams on (trace height = amplitude multiplier)



MODEL VALIDITY: WARNING
P_up=0.9876 P_n0=0.5306 P_target=0.5226 (P_up*P_n0=0.5240)
populations are per trapped atom: trapped 0.5202 (lost 0.4798); of all atoms: P_up=0.5138 P_n0=0.2760 P_target=0.2718
nbar (trapped atoms) 1.45 -> 0.9717 T_equiv=4.06 uK [Energy-matched mixture of local bound canonical distributions, weighted by surviving spatial populations; proxy only. None means no finite positive-temperature match.]
Well-loss fraction, assuming no recapture: 0.4798
Axis temperature proxies (uK): x=6.52, y=7, z=4.06
photons/atom: pump 2.42, repump 2, Raman 0.000551
target $|3,3\rangle$ scattering rate 28.4 /s; quanta removed per photon None
t(P_up>=0.9) = 2.1 ms; t(P_target>=0.5) = 3.67 ms
conservation err 4.3e-14; min eig 0; boundary pop unspecified; overflow 0.48
solver coherent_3d_split_ensemble, wall 293.8 s

[OK] conservation: max |survival + well loss - 1| = 4.29e-14
[OK] positivity: minimum density eigenvalue / population = 0
[OK] weak excitation: maximum eliminated excited fraction across nodes and spin states = 0.000617
[WARNING] motional secular: max optical rate / smallest local motional angular spacing across all axes = 0.48
[WARNING] carrier calibration: Synthetic measured-envelope fit RMS = 0.174; residual gamma = 0.000682/s. A poor fit is an unresolved calibration-model mismatch, not a validated intensity estimate.
[WARNING] three-dimensional recoil: Joint spin/nx/ny/nz populations retained, but emission uses resolved spherical dipole channels with factorized Cartesian marginals. Interference between emitted spherical channels and correlations of one photon's Cartesian recoil components are neglected.
[WARNING] recoil coherence: Population-resolving scattering erases motional coherence and treats elastic spin amplitudes as distinguishable; this is an additional approximation beyond secular averaging.
[WARNING] lattice: Separable isolated wells with Gaussian local depths; no tunnelling, coherent Raman continuum excitation, envelope transport or recapture. Scalar trap basis with diagonal differential shifts; trap shape changes from spin-dependent shifts neglected.
[WARNING] Stark shift approximation: Raman/trap Stark operators use diagonal matrix elements; static off-diagonal motional dressing and trap-basis changes are omitted. Excited-state lattice shifts of the pumping transitions remain the configured common optical shift.
[WARNING] spatial ensemble: 64 frozen Gaussian position / uniform registration samples. Refine samples and time_step_us for convergence; samples are numerical quadrature, not experimental error bars.
[WARNING] temperature: Axis temperatures are energy-matched bound-state mixture proxies; transverse survivor cooling may result from preferential loss.